ReaATS

tjingboem productions

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What it does

ReaATS is a partial editor for REAPER, in the spirit of SPEAR (https://www.klingbeil.com/spear/). It opens ATS analysis files, and shows the sound inside them: a set of sine waves (the partials) that each follow their own frequency and loudness through time, plus the noise that is left over. You select parts of the partials, delete them, transpose them or move them, listen to the result, and put the result in your project as a new sample. ReaATS reads and writes standard ATS files, the ones that Csound's ATS opcodes use.

An ATS file is not a recording. A sound is analysed into partials, the sine waves that make up its pitched part, and a residual, the rest, stored as the energy in 25 bands. ATS can build the sound back from these two things. Because every partial is a separate line, you can change one harmonic of a saxophone and leave the rest alone, which you cannot do in an ordinary waveform.

ReaATS is a Lua script that runs inside REAPER. It makes the sound itself, so it does not need any other program to play or print an edited sound.



What you need and installing

- REAPER, with the script loaded from the action list.
- The SWS extension, for AUDITION. PRINT does not need it.
- An .ats file. A small example, sax.ats, comes with ReaATS in the examples folder. To make your own you need Python 3 with numpy, for the analysis program that comes with ReaATS (see the Analysis page, below).

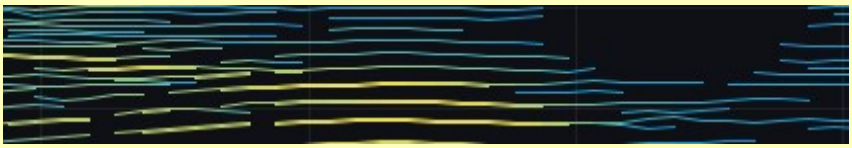
On a Mac, REAPER reports its Ctrl shortcuts on the Cmd key, so wherever this manual says Ctrl, use Cmd (and Option where it says Alt). Temporary files (the sounds of AUDITION, the edits handed to the Python program) go to a folder named ReaATS_temp inside REAPER's own resource folder, which REAPER chooses on every system; they are removed when the editor closes. Before a very large analysis is loaded, ReaATS asks the Python program how much memory the computer has free, so that it can refuse an analysis that would not fit.

Put the ReaATS folder (ATS_Editor.lua and the lib folder next to it) in REAPER's Scripts folder, for example Scripts/tjingboem/ReaATS. In REAPER open Actions, Show action list, choose ReaScript: Load and pick ATS_Editor.lua. Run it from the list, or give it a shortcut or a toolbar button.

ReaATS starts with an empty window: it does not reopen the file you used last. The lib folder must stay next to the script: it holds the reader and writer for .ats files and the editing logic.

Python. The analysis is done by a small program that comes with ReaATS, python/reaats_analyse.py. It needs Python 3.8 or newer and the numpy package on your computer, and nothing else. REAPER users usually have Python already. ReaATS looks for it in the place REAPER is set to use for ReaScript (Options, Preferences, Plug-ins, ReaScript), and then in the usual places. This is checked once, automatically, the very first time ReaATS is run, with a popup saying either way: Python found (and which numpy), or what to do if it was not. Once Python has been found working, it is not checked again automatically; a failure is not remembered, so it is checked again next time, in case Python gets installed in between. When no Python is found, the Analysis page shows a red warning with a Help button for the same popup.

If ReaATS cannot find a Python that will do, a window opens (the first time ReaATS runs, and again whenever you click ANALYSE with nothing found). It says what is missing: no Python, no numpy, or a Python that is too old. It tells you to download Python from python.org and install it, to install numpy (python3 -m pip install numpy) and to tell REAPER where Python is, in Options, Preferences, Plug-ins, ReaScript. It also offers to open the download page. After you have done this, open the Analysis page again, or restart REAPER, and try again. **There is no analysis without Python.**



The window, looking around and selecting

The window

The picture is time from left to right and frequency from bottom to top, on a logarithmic scale, so every octave has the same height. The Hz labels are on the left and the note names on the right. At first the labels show only the C's, and zooming in shows more notes.

- Every sine partial is a line. The brighter the line, the louder the partial at that moment. Partials more than 80 dB below the loudest are not drawn, but they stay in the file.
- The faint orange layer behind the lines is the noise, shown in its 25 bands. The checkbox Noise layer, at the top right, switches it on and off. The checkbox Sine layer, just under it, does the same for the partial lines. The two layers can be shown alone or together. The same switches decide what you hear in AUDITION and what PRINT makes: with Sine layer off you hear only the noise, and with Noise layer off only the sines. While the sine layer is off you cannot select partials, so that no edit is made on something that is hidden.
- The line at the bottom left gives the file name, the number of partials and frames, the length, the sample rate and the distance between frames. A star after the name means there are changes that have not been saved. For a few seconds after an action, a green message that says what it did takes the place of the numbers (the name and the star stay).

  fox47.ats | 190 partials 555 frames 2.76 s 44100 Hz hop 5.0 ms type 4 (phase) (noise)

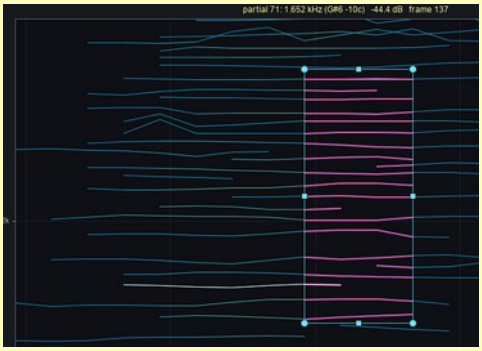
- While you drag, a crosshair shows the time, the frequency and the note name. When the mouse is close to a partial, that partial turns white and its number, frequency and level are shown at the top.
- Selected points are pink.
- The playhead is a yellow line across the picture, with a marker on the time ruler and its time in a small label. It shows where AUDITION and PRINT start, and it moves while the sound plays. Home puts it at the start (the small |< button under PRINT does the same).

- The buttons are all in the top row. From the left: ANALYSE, EXPORT .ats, MAX PARTIALS, the SPACE controls, then PRINT, AUDITION and the two layer checkboxes at the right. The selected points are summed up at the bottom right.



Looking around

Control	What it does
Mouse wheel	Zoom in time, around the mouse.
Ctrl + mouse wheel	Zoom in frequency, around the mouse.
Shift + mouse wheel	Scroll in time.
Right or middle mouse button, drag	Pan in time and frequency.
Right-click (no drag)	Without a selection: show the whole file, like F. With a selection: open the gain slider (see Gain of the selection, under Editing). With a stretch waiting in the box (dragged, not applied yet): open the slant slider instead (see The slant of a stretch, under Editing). On a border dot of the bend box: remove the dot.
F or Home	Fit the view to the whole file.
P	Show or hide the partials.
N	Show or hide the noise layer (the same as the checkbox).
Z	Zoom to the selection, with some room round it (a quarter of its size at each side, at least a little). Without a selection nothing happens and the status line says so.



Selecting

Drag with the left mouse button to draw a box. Every point of every partial inside the box is selected. Hold Shift while you start the box to **add** to the selection. A click on empty space clears the selection, and so does Esc.

A selection is made of points: one partial at one moment (one frame). This means you can select a piece of a partial and leave the rest. The header line shows how many points and partials are selected and the time and frequency range they cover.

Lasso. Hold Ctrl and Shift and drag to draw a freehand loop. When you let go, the loop is closed and the points inside it are selected, in every channel that is on and green, like a box. A lasso always **adds** to the selection you have; to start over, press Esc or click on empty space first (with nothing selected, a lasso is simply a new selection). A lasso follows a curved line of partials much better than a box.

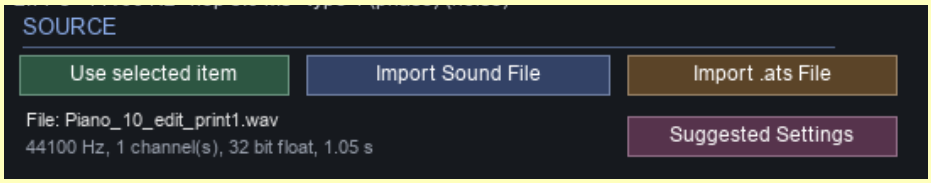
Select all. Ctrl+A (Cmd+A on a Mac) selects the whole sound: every point that is drawn, over the whole length of the file and the whole frequency range of what sounds, in every channel that is on and green. The box is made round all of it, so that you can drag its dots and edges at once, for example to stretch the whole sound. Points that are quieter than the display threshold are not drawn and are not selected, as with a box. The view does not change: Z zooms to the selection and F fits the whole sound. Esc clears the selection.

Making an .ats file: the Analysis page

The ANALYSE button opens the Analysis page. With no file open yet, it is a bigger button centred above the empty picture, labelled "ANALYSE or IMPORT FILE" to make clear the page is also where an existing .ats file is opened, not only where a new sound gets analysed; once a file is open, it moves to its usual small spot at the top right, labelled plainly ANALYSE. Esc, or the EDITOR button, returns to the editor.

Importing an .ats file. The button Import .ats File on the page opens an existing .ats file, for example one that you made with Csound or SPEAR, and returns to the editor. It does the same as the O key in the editor.

Choosing the sound. Select an audio item in REAPER and click Use selected item. Only the part of the file that the item plays is analysed, from its start offset for its length. Volume, fades, effects and pitch changes on the item are not applied, because the analysis reads the file itself. If you want that, render the file first. Click Import Sound File to pick a sound file: the file dialog shows only .wav, .flac, .aif and .aiff files. The sound can be a WAV file (PCM, 8 to 32 bit, or float), an AIFF file (.aif, .aiff, also AIFC with uncompressed or float samples) or a FLAC file (8 to 24 bit). ReaATS reads them itself, nothing else has to be installed. WAV and AIFF are read at once. A FLAC file is decoded first, in the analysis program, at about half a million samples per second: a minute of stereo sound takes about 10 seconds and nine minutes of stereo about 100 seconds, which comes before the analysis itself; the page then says "reading the sound". If the item's file is another format (mp3, ogg, video and so on), render it to WAV first (Glue, or File, Render) and select the new item. The page shows the sample rate, the channels and the length of the sound, in seconds and, for a minute or more, also as minutes:seconds (536 seconds is 8:56).

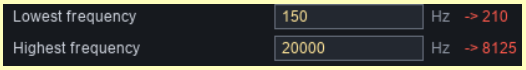


Looking at the sound first. The button Suggested Settings, under the file name, takes a quick look at the sound before you analyse it (a second or so, however long the sound is; a long sound is sampled). It opens a view with:

- the **average spectrum** on a log scale, with the **strong low partials** marked and named (for example "C4 261"), and the highest frequency that has content ("content up to 17331 Hz");
- the **loudness over time**;
- your current **lowest and highest frequency** drawn as white lines, and the **suggested** ones in red;
- a list of **suggested settings**, each with the reason: the lowest frequency (just below the lowest strong partial, so that the window is short enough for the attack), the highest frequency (just above the content), the amplitude threshold (-60 dB for a short sound, higher for long ones), the hop size (shorter for a struck sound) and the window cycles.



The button Use all suggestions uses them all. Back to the settings (or Esc) returns to the page, where the suggested values stand in **red** to the right of the fields they belong to, with an arrow, for example "-> 210". Point at a red value to read why; click it to use it. A red arrow disappears when you use its suggestion. Instead, the value in the field itself turns **red** if the suggestion changed it, so that you can see at a glance what the suggestions did to your settings (a suggestion that was the same as your value changes nothing and shows nothing). A value stays red until you change that field yourself (by typing, with the mouse wheel or with a menu), press Defaults or choose another sound. The suggestions are a starting point, not a rule: the lowest strong partial is the lowest peak that stands out and is within 25 dB of the strongest low peak, so a note whose fundamental is very weak, or a sound with a loud low rumble, may need another value; the peaks the view names help you judge.



The options. The page has every option of the analysis. Click a value to type a new one in the field itself: all of the old value is selected, so the first key you type replaces it. Enter accepts the value, Tab accepts it and goes on to the next number, Esc cancels, and a click anywhere else accepts it. Or point at a value and use the mouse wheel (Shift for finer steps). Point at an option to see a short explanation at the bottom. The line under the options shows what the settings mean for this sound: the window length in samples and milliseconds, the FFT size, the hop and the number of frames.

Option	What it sets	Default
Start, Duration	Where to start and how long to analyse, in seconds. 0 means the whole sound.	0
Channels to analyse	How many channels are analysed, each into a file of its own (see below). When you choose a sound, this is set to all its channels.	all
Mix all channels to mono first	Analyse the average of all channels as one sound.	off
CPU cores	How many channels are analysed at the same time, one per core. 0 is automatic: all cores but two, so that REAPER keeps room to play, and at least one. Next to the box the page shows how many run at once. It does nothing for a single channel.	0
Lowest frequency	The analysis window is four cycles of this frequency long, so a low value gives a long window: good frequency detail, but fast changes are smeared in time. Set it a little below the lowest note in the sound (Suggested Settings shows where that is).	20 Hz
Highest frequency	The highest frequency analysed.	20000 Hz
Window cycles	How many periods of the lowest frequency fit in the window.	4
Window type, alpha, beta	The FFT window and its parameters.	blackman-harris-4-1
Hop size	The distance between frames, as a fraction of the window.	0.25
FFT size, Window length	Force a value instead of the automatic one.	automatic
Amplitude threshold	The quietest peak that counts, in dB below full scale. Raise it for noisy sounds and for long sounds (see Long sounds: choosing the amplitude threshold).	-60 dB
Frequency deviation	The largest relative frequency change between two frames when peaks are matched into a track.	0.1
Track length, Minimum segment length, Minimum gap length	How tracks are smoothed, how short a piece may be, and how long a gap may be bridged, in frames.	3
Last peak contribution, SMR contribution	How much the previous frame and the signal-to-mask ratio count when peaks are matched.	0, 0
Optimise, Optimise threshold	Prune quiet and short tracks afterwards, with an extra threshold if wanted.	on
Store the phase, Store the noise bands	What goes into the file. Without noise bands the analysis is faster.	both on
Share rows between tracks	Tracks that do not overlap in time use the same partial row of the file, with one silent frame between them. The sound is exactly the same, and the file of a long sound is 20 to 100 times smaller: 240 seconds of music gave 8894 rows and a file of 4.9 GB, and about 100 rows and a file of some tens of MB when rows are shared. Switch it off only if you want every track to have a row of its own, as the original ATS analysis does. A row then holds several unrelated tracks one after another, so a row is no longer "one partial".	on

The Defaults button resets the options. Your settings are remembered the next time.

Analysing. Click ANALYSE. The analysis runs in the background while the page shows it at work: the page is dimmed, the big word ANALYZING... blinks, the ANALYSE button blinks, and a bar shows how far the analysis is, with the percentage and the seconds so far. Cancel stops the analysis and leaves no file behind. If the analysis program has stopped answering, for example because it crashed, the page notices after two minutes without a sign of life and says so, so the page never waits forever. A Cancel that the program does not answer ends after 8 seconds. When the analysis is done the editor is shown at once, with the first channel open and the time each step took in the status line. The ANALYSE button shows the page again with the list of files that were made. You can also leave the page with Esc while it works: the analysis goes on, and the result opens in the editor when it is done. The page then shows what was made, and the first file is open in the editor. The result is saved in the ReaATS folder of the project, as name.ats (see Where the files go). If the project has not been saved yet, it is saved next to the sound. If that name exists, name_2.ats is used. With Change... you choose another place and name, and ReaATS asks before it replaces a file.

Several channels. With Channels to analyse set to 2 or more, each channel is analysed on its own with the same options and written to its own file: name_ch1.ats, name_ch2.ats, and so on. All the files of a set have the same frame size, number of frames and length. The channels are analysed at the same time on separate cores of your computer, so a set of channels takes much less time than one channel after the other: in a test, 8 channels of 20 seconds took 11 seconds instead of 30. The result is exactly the same files as when the channels are analysed one after the other. The files only get their real names when every channel is done, so a cancelled analysis leaves nothing behind and does not touch a file that was there before. The editor opens the first channel. The editor opens the first channel by itself, and with it the whole set. To open one of them again later, use Import .ats File or the O key.

The program also works from a terminal, for scripts:

python3 python/reaats_analyse.py sound.wav sound.ats --lowest-frequency 100

python3 python/reaats_analyse.py --help lists all the options. The analysis is the one of the pyatsyn package. Csound's own analyser, csound -U atsa sound.wav sound.ats -l 100, makes files that ReaATS can open too, and so do ATSA and ATSH. If the analysis sounds wrong, try a different lowest frequency first. For a sound whose lowest note is above 100 Hz, try 100. The time an analysis takes depends mostly on how many peaks it finds, not on how long the sound is. A long or noisy recording with the default threshold of -60 dB finds thousands of tiny partials, most of them noise, and can take hours. In one test, the first 30 seconds of a 9 minute recording took 52 seconds at -60 dB, 18 seconds at -50 dB, 10 seconds at -40 dB and 5 seconds at -30 dB. So for long sounds, music and field recordings, raise the amplitude threshold to about -40 dB, or analyse a shorter part with Start and Duration. The page shows a warning when a long sound is set to a low threshold. The noise that the many tiny partials would have caught is stored in the noise bands anyway.

Settings that put sound into the noise. Whatever the analysis does not follow as a partial is left over in the noise bands. So a high amplitude threshold (for example -40 dB) leaves out every partial quieter than that, and a low Highest frequency (for example 5000 Hz) leaves out everything above it: all of that ends up in the noise, and the resynthesized noise then sounds much too loud and hissy, in a piano note for example, where the quiet upper partials carry much of the character. The page remembers its settings from one sound to the next, so a threshold that you set for a long recording is still there when you analyse a short note. For a short tonal sound use -60 dB or lower and a Highest frequency of 16000 to 20000 Hz.

Choosing the analysis values: advantages and disadvantages

Every setting on the Analysis page moves sound between two representations, sine partials and noise bands, or trades one kind of accuracy for another. No setting is correct in itself. This chapter describes, for each important setting, what it changes, what is gained by raising or lowering it, and what is lost. The last sections give measurements and a procedure for testing.

The principle: partials and noise divide the sound

The analysis finds spectral peaks frame by frame and joins them into tracks, the sine partials. What is not followed as a partial is not discarded: it remains in the residual, which is stored as 25 noise bands. A setting that makes the analysis follow fewer peaks (a higher threshold, a lower Highest frequency, a longer Minimum

segment length) therefore does not remove sound. It moves sound from the partials, which can be seen, selected and edited, to the noise, which can only be heard and shaped as bands. The cost of that move is that the sound is described less precisely. The gain is a smaller number of partials, a faster analysis, a smaller file and a picture that can be worked with.

Two properties of the threshold are often misread:

- It is **absolute**. -60 dB means a sine with an amplitude of 0.001, 60 dB below full scale. A recording whose loudest part is at -30 dB is covered over only 30 dB by the same setting. Suggested Settings reports the loudest value; the range actually covered is that value minus the threshold.
- A **lower** threshold is a **more sensitive** analysis: -60 dB is more sensitive than -40 dB.

Amplitude threshold

Effect	The quietest peak that is followed as a partial.
Lower (e.g. -70 dB)	**Advantages:** quiet partials, decays and the upper partials of an instrument stay editable sines; the noise layer is small and quiet; the highest fidelity to the original. **Disadvantages:** many more partials (see the measurements below), a much longer analysis, larger files, slower drawing and editing; many of the additional partials are short and unstable, which makes the picture hard to read.
Higher (e.g. -40 dB)	**Advantages:** a clean picture of the structurally important partials; a fast analysis; small files; practical to edit by hand. **Disadvantages:** everything quieter is now noise. The noise layer is louder and, because it is described by bands, it has no pitch structure: a harmonic tail may turn into hiss.
Suitable when	Low: short sounds (a note, a phrase), where each partial counts. High: long recordings, mixtures, and any case in which the aim is editing the main lines instead of reproducing the sound.

Highest frequency

Effect	Peaks above this frequency are not followed as partials. They remain in the noise bands.
Lower (e.g. 5000 Hz)	**Advantages:** fewer partials and a faster analysis; the upper range of most instruments consists of weak, closely spaced components that are in any case hard to edit. **Disadvantages:** a region of the spectrum that does contain harmonic structure (the upper partials of a piano or a bright brass tone) becomes noise, and its pitch relation to the fundamental is lost.
Higher (up to 20000 Hz)	**Advantages:** the full bandwidth stays available as sines, which matters for bright sounds and for transposition, where noise does not follow the pitch. **Disadvantages:** many additional weak peaks in a region where the sound is often noise-like to begin with.

When high partials can be treated as noise

The two settings above interact. The upper partials of most instruments are 40 dB or more below the strongest component. With an amplitude threshold of -40 dB they are therefore, in practice, already outside the sines, and a high Highest frequency adds nothing to the analysis except the few loud peaks that exceed the threshold. In this situation the high region is better described as noise: the ear does not resolve closely spaced partials above a few kilohertz as separate pitches, and the noise bands (critical bands) follow its resolution. The consequence is a rule of thumb. When the threshold is high, Highest frequency can be lowered without a noticeable loss, and the analysis is faster and the file smaller. When the threshold is low, the opposite holds: the quiet upper partials are exactly what the low threshold is chosen to keep, and limiting the bandwidth would discard them again.

The limits are shown by a counter-example. A piano note analysed with a threshold of -40 dB and a Highest frequency of 5000 Hz sounded wrong. The noise was loud and hissy, and in the decay of the note the sine partials were 1.4 to 9 dB below the noise. The quiet upper partials of a piano carry a good part of its character, and all of them had become noise. For a sound with a clear harmonic spectrum and few partials, the high setting is not appropriate.

Lowest frequency and window cycles

Effect	The analysis window is window cycles periods of the lowest frequency long. A lower Lowest frequency or more cycles give a longer window.
Longer window	**Advantages:** finer frequency resolution; closely spaced partials, and the low partials of a harmonic sound, are separated. **Disadvantages:** worse time resolution; attacks are smeared, and fast changes of amplitude or pitch are averaged over the window. The window is longer than necessary if Lowest frequency is set far below the lowest pitch that occurs.
Shorter window	**Advantages:** sharp attacks and quick changes are followed. **Disadvantages:** neighbouring partials may merge into one peak, and low partials are lost or unstable.
Note	The default of 20 Hz gives a very long window and a poor result for most material. Setting Lowest frequency a little below the lowest fundamental that occurs (100 Hz for an alto saxophone) is usually the better choice.

Window type

The window determines how much a strong peak leaks into its neighbours. Blackman-Harris (the default) suppresses leakage strongly, so that weak partials next to strong ones can be found, at the price of a wider main lobe, which makes the resolution of two close partials slightly coarser. Hann and Hamming have a narrower main lobe and higher side lobes: closer partials are resolved, but weak ones next to strong ones are masked. The rectangular window gives the highest leakage and is rarely a good choice. The differences are small compared with those of the threshold and the window length; the default is the safe choice.

Hop size

Effect	The distance between two frames, as a fraction of the window.
Smaller (e.g. 0.125)	**Advantages:** more frames; faster changes are followed, and tracks are matched more reliably because a partial moves less between frames. **Disadvantages:** about twice as many frames, so a longer analysis, a larger file and more data to edit.
Larger (e.g. 0.25)	**Advantages:** a lighter analysis and file. **Disadvantages:** attacks and quick modulations are followed less closely; with a large hop the tracks are more likely to be broken or confused.

Frequency deviation

The largest relative change of frequency between two frames at which peaks are still joined into one track. A small value keeps the tracks stable and prevents unrelated peaks from being joined, but cuts glissandi and wide vibrato into short pieces. A large value follows fast pitch movement, but may connect two different partials, so that a track jumps between them. With a small hop size a smaller deviation is sufficient, because a partial moves less between two frames.

Minimum segment length, minimum gap length, track length

These settings clean the tracks after the peaks have been found.

- **Minimum segment length** (a higher value): removes short, flickering tracks. The result is cleaner and the file smaller; short events, such as the very beginning of an attack, may be lost. A lower value keeps them and keeps the clutter.
- **Minimum gap length** (a higher value): a track that disappears for a longer time is treated as two. A lower value bridges only short interruptions, so vibrato or tremolo is not joined into one long track where the partial is in fact absent.
- **Track length**: the number of frames over which the frequency of a track is smoothed. Longer smoothing gives steadier pitches and removes analysis jitter, but also rounds off real, fast pitch changes.

These settings do not make the analysis itself faster. They change what is kept afterwards.

Settings that change the file and not the sound

- **Share rows between tracks** (on by default): tracks that do not overlap in time use the same partial row. The sound is the same (the synthesis is identical to within rounding), and the file of a long sound is 20 to 100 times smaller. The only disadvantage is that a row no longer corresponds to one continuous partial, which matters to anyone who reads the file with other software in the expectation that it does. Csound plays such files without difficulty.
- **Store the phase**: AUDITION and PRINT play every partial with the phase that the analysis measured, so the waveform of the original is rebuilt: a firmer sound, with sharper attacks and more body in the bass. Without the phase the file is smaller, but every partial gets its phase from its frequency alone, and the sound is softer. Pieces of partials that you change are played with phases that follow their new frequencies.
- **Store the noise bands**: without them the analysis is faster and the file smaller, but the noise layer is lost completely. The sound is then sines only, and everything below the threshold is gone.

Summary of the trade-offs

Setting	Raise it	Lower it
Amplitude threshold	Fewer partials, faster, smaller; more sound in the noise	More partials, slower, larger; more detail kept as sines
Highest frequency	More bandwidth as sines, more partials	Fewer partials; upper range becomes noise
Lowest frequency	Shorter window: better time resolution, poorer low frequencies	Longer window: finer frequency resolution, smeared attacks
Window cycles	Longer window	Shorter window
Hop size	Fewer frames, coarser in time	More frames, finer in time, larger file
Frequency deviation	Follows glides, risk of wrong joins	Stable tracks, glides are cut
Minimum segment length	Cleaner, loses short events	Keeps short events and clutter

Measurements: how the cost follows the number of partials

A recording of 536 seconds (stereo, 103 MB) analysed with the default threshold of -60 dB needed more than 1000 seconds to reach 37 percent. To find out why, the first 30 seconds were analysed four times with different thresholds (Start and Duration), and the time and the number of partials were recorded:

Threshold	Time for 30 s	Partials found
-60 dB	52 s	7705
-50 dB	18 s	2384
-40 dB	10 s	716
-30 dB	5.3 s	121

Each step of 10 dB made the analysis three to six times faster, and the time followed the number of partials, not the length of the sound. In a music recording there are a few loud partials and, below them, a very large number of quiet peaks (note tails, reverberation, room noise, the upper partials of everything), and each of them must be followed from frame to frame. The numbers are from one recording; the proportions, about three times more partials for every 10 dB, are the part that carries over to other material.

Two further consequences follow. The work per frame grows with the number of partials alive at the same moment, so the progress of a long analysis is not linear and a percentage reached early is no basis for extrapolation. And the size of the file and the time needed to draw, select and edit grow with the number of partials as well. Without row sharing, 240 seconds of music gave 8894 rows and a file of 4.9 GB for one channel. With Share rows between tracks, a 60 second test went from 1422 rows to 11, and from 206 MB to 2.7 MB. Row sharing solves the file size, but not the analysis time, and not the practical problem that thousands of faint lines cannot be edited by hand.

Starting values by length of the sound

These are also the values used by the red suggestions in Suggested Settings. They depend on the length of the part that is analysed.

Length	Threshold	Reasoning
Up to 2 minutes	-60 dB	Few partials; the analysis takes seconds to a few minutes; the quiet partials carry the character of the sound.
2 to 10 minutes	-50 dB	About three times fewer partials than at -60 dB, with the loudest 50 dB still followed.
Longer than 10 minutes	-40 dB	At this length even -50 dB gives a very large number of partials and a long analysis. The 536 second recording finished in about 15 minutes at -40 dB, most of it writing a very large file, which row sharing has since solved.

The page remembers its settings from one sound to the next. A threshold chosen for a long recording is still set when a short note is opened, and makes the analysis of that note too coarse. Check the threshold whenever a new sound is chosen, or press Defaults.

A procedure for deciding

- **Look at the sound.** In Suggested Settings, read the loudest value, the dynamic range ("the quietest tenth is ... dB below"), the loudness curve and the spectrum, where the content stops.
- **Decide the purpose.** Editing single partials, or rebuilding the sound from a few clear lines, favours a high threshold. Faithful reproduction of the whole sound favours a low one, which is practical for short sounds only.
- **Test on a short part.** Set Start and Duration to cover 20 to 30 seconds that are typical of the whole, and try two or three thresholds, for example -60, -50 and -40 dB. Record the time and file size of each. The time for the whole sound is roughly the test time multiplied by the ratio of the lengths, and somewhat more because the progress is not linear.
- **Listen to each result.** Switch the noise layer off (N) and audition (A): this is the sines alone. Then switch it on. If the sines carry what is wanted and the noise is quiet and in the background, the threshold is high enough. If the noise is loud and hissy, or the decays have disappeared, the threshold was too high.

- **Choose the highest threshold that still sounds right.** Every 10 dB is a factor of about three in time, size and work.
- **If this is still too slow, analyse parts.** Start and Duration cut out the section that is needed. The result is a smaller file that keeps all the detail that was asked for.

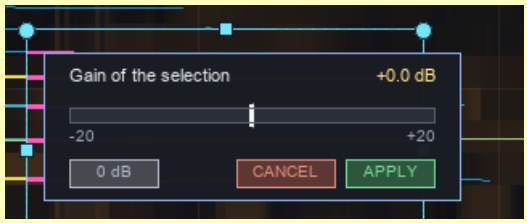
Editing

All edits work on the selected points.

- **Delete** makes the points silent. Their amplitude becomes 0, which is how ATS marks a partial that is not sounding.
- **Up and Down** transpose the points by a semitone. Shift makes it 10 cents and Ctrl makes it an octave.
- **Dragging** a selected point moves the whole selection. Press on one of the pink points to start. Sideways movement shifts the points in time, in whole frames, because the file stores one value per frame. Vertical movement changes the frequency continuously. While you drag, the selection dims and a lighter copy shows where it will land. Hold Ctrl to move in one direction only, the one you move most: the time stays the same, or the frequency stays the same. You can press or release Ctrl in the middle of the drag. Hold Shift for fine movement, a fifth of the mouse movement, for precise placing; Shift can also be pressed and released during the drag. For the same reason there are the arrow keys, which move in time only (Left and Right) or in frequency only (Up and Down).
- **The crosshair.** While you drag something (a move, a box or a lasso), thin lines cross the picture through the pointer, with a small label: the time and frequency of the pointer, or for a move how many frames and semitones the points will move. For a move the lines show where the points really go, a faint cross marks where the drag began, and when Ctrl keeps one axis, the line of that axis is drawn in colour and the label says "time only" or "frequency only".
- **Left and Right** move the selection one frame in time. Ctrl makes it ten frames.
- **Ctrl+C, Ctrl+X and Ctrl+V** copy, cut and paste, see below.
- **Ctrl+Z** undoes the last edit, and **Ctrl+Y** redoes it. Ctrl+Shift+Z redoes too.
-

A piece of a partial that moves keeps its own row in the file if the place it moves to is empty. If that row is busy there, the piece goes into the nearest row that is silent, and if there is none it gets a new row. Nothing is ever written over other data, so two partials can sit at the same time and frequency and both sound. A move cannot take points outside the first and last frame of the file. A file can therefore end up with more partials than the original.

After a move or a transposition the selection stays with the points, so you can go on with the same selection.

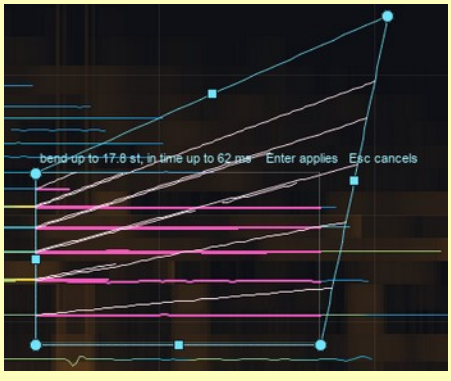


Gain of the selection. With points selected, a right-click (without dragging) opens a small slider at the pointer, from -20 dB to +20 dB. Drag it, or press Up and Down for half a dB; it sticks at 0 dB, and the 0 dB button puts it back there. Enter or APPLY multiplies the amplitude of the selected points by that many dB, in every channel that is on and green, as one undo step; Esc, CANCEL or a click outside leaves everything as it was. Only the sines change: the noise bands are not touched. While the slider is open it takes every click and key. If the sound ends up louder than -1 dBFS, AUDITION and PRINT turn it down (see The level).

Copy, cut and paste. Select points and press Ctrl+C to copy them. Ctrl+X copies and then deletes them. Put the playhead where you want the copy (click in the picture or on the time ruler) and press Ctrl+V: the points are pasted so that the first copied point lands at the playhead, with the same frequencies and amplitudes. You can paste as often as you like, at other places, and in a file you open later in the same editor window. The pasted points are selected, so you can move them, transpose them or bend them right away. A paste works like a move: it never writes over other sound. A piece goes into the same row when the place is empty, otherwise into the nearest silent row, otherwise into a new row. Points that would land after the end of the file are left out, and the status line says how many. With several channels, a copy takes the selected points of every channel that is on and green, and a paste puts every channel's points back into the channel with the same number, if that channel is on and green now. The channels keep their timing against each other. One Ctrl+Z takes a paste back from all the channels at once.

The box and its dots: bending and stretching the selection. The selection has a box round it: the box you drew, or for a lasso the box round the loop. There is a dot in each corner of the box, and the selected points behave as if they were printed on a rubber sheet that you can bend by dragging the dots. Everything follows the sheet smoothly, so a selection can be bent, tilted, stretched or squeezed with one gesture. Up and down is frequency, and every partial keeps its harmonic relationship, because the points move by a ratio (a number of semitones), not by a number of Hz. Left and right is time: the points move later or earlier.

- Drag a **corner dot** up or down to bend in frequency, sideways to move in time, or both at once. The upper right corner up bends the end of the selection up, and the upper points more than the lower ones.
- Drag the **square handle in the middle of an edge** to move the whole edge (you can also drag the line itself, but see below). Dragging a selected point that happens to lie on the line moves the selection instead of the edge, so the square in the middle is the sure way to grab an edge. The right edge up is a plain pitch bend: everything at the right end is raised, and the points further left less, in a straight line. The right edge to the right is a **time stretch**: the selection gets longer, and everything in it slows down. The same edge to the left squeezes it. The top edge up stretches the frequencies (the higher the point, the more it moves), and the top edge sideways shears the sound, higher partials later than the lower ones.
- **Ctrl+click** on the line of an edge, away from the square handle in the middle, adds a **border dot**, which you can drag in any direction. With dots you can make curved bends and uneven stretches: a glide that eases in, a vibrato, a stretch that is slow at first and fast at the end. **Right-click** on a border dot removes it. **Ctrl+Shift+drag** on an edge between two dots arches it like a curve, up and down or sideways, whichever you drag most.
- **Shift** while dragging makes the movement a fifth as large, for fine adjustment. **Ctrl** while dragging keeps one direction only (the one you move most): only frequency, or only time. You can hold Ctrl from the moment you press on a dot or on the square handle of an edge, or press it during the drag. A sideways drag with Ctrl held cannot add a pitch bend by accident. A label shows the bend in semitones and milliseconds, and the points drawn in lighter colour show where they will go.
- **The picture follows the drag.** When the pointer comes into the last stretch of the picture (the outer eighth on the right or on the left) while you drag a dot or an edge, the picture scrolls on, faster the further in the pointer is, and the dragged dot stays under the pointer. Past the end of the file there is empty space to scroll into, so you can drag a stretch as long as you like, and the file gets longer when you press Enter. Move the pointer back from the edge to stop the scrolling.
- **Time cannot fold back.** A movement that would make a partial go back in time is not made: the dot waits at the last place where it was possible, and you can drag back the other way.
- **Enter applies** the bend, as one undo step in every channel that is on and green. **Esc** or **Ctrl+R** put the box back to flat without changing anything. If you start another selection or another edit, or press AUDITION, PRINT or save, a bend that was shaped is applied first, so it is never lost; Ctrl+Z takes it back.
- After Enter the box stays, flat again, round the points where they are now, so you can continue.



The slant of a stretch. You decide how far to stretch by dragging the box. To spread the stretch unevenly over the selection, so that it stretches less at the left border or less at the right border, right-click while the stretch is waiting (dragged, but not yet applied with Enter). The slant slider opens in place of the gain slider. Its left end is "less at left": the left border keeps its speed and the stretch builds up towards the right. Its right end is "less at right", the other way round, and the middle is an even stretch (the slider sticks there). A line

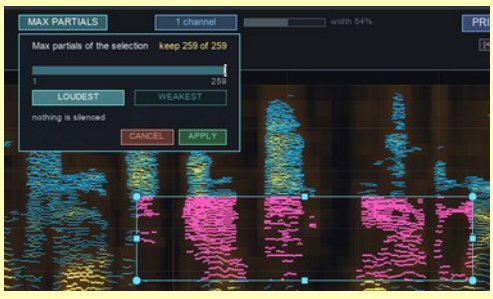
shows the speed at both borders, for example "left border x1.00 right border x3.00", and the box shows the result while you drag. APPLY applies the stretch, as Enter does. CANCEL, or Esc, gives the box back as it was before the slider, with the stretch still waiting. EVEN puts the slider in the middle. The right border ends where you dragged it: only the way the time is spread over the selection changes. A squeeze that would fold time back on itself is limited automatically. To keep the attack of a sound near its own speed and slow down the part that follows, select the sound, drag the right edge to stretch it, and slide the slant towards "less at left".

How a change in time is made: the points of a partial sit on the frames of the file, one value per frame. Every piece of a partial that you moved or stretched is made again on the frames that lie between its new first and last time, from its own points by straight lines, for the amplitude and for the frequency. A stretched piece therefore gets more frames and a squeezed one fewer. As with a move and a paste, nothing is written over other sound: a piece stays in its row if the frames it needs are empty, otherwise it goes to the nearest silent row, otherwise to a new row. If a stretch runs past the end of the file, **the file gets longer**: silent frames are added after the last one, in every channel of a set (they must stay the same length), and the pieces use them. The status line says how much longer the file is now, the length in the header grows, and the saved file is longer too. One Ctrl+Z makes the file short again in all the channels, together with the stretch. A file can grow as far as the memory allows. The limit is the number of new values: a sound with few partials can be stretched to hours, one with thousands of partials can grow by a few seconds. If a stretch goes further, the part that does not fit is cut off and the status line says how far the file can grow. A stretch can also go **before the start** of the file: drag the left edge or a left corner to the left of 0 (the picture follows the drag there too, and what lies before the start and after the end of the file is drawn darker). When you press Enter, silent frames are put **in front** of the file and the stretched pieces use them. The file has to start at 0, so **everything else in the file moves later** by the same amount, in every channel: the unselected sound, the noise, the playhead and the picture, so that nothing seems to jump. The status line says how many frames were put in front and how much later everything is now. One Ctrl+Z takes the frames away again and puts everything back. What lies beyond the room that is allowed is left out; the status line says if a whole piece fell outside. The same holds for a **move** and a **paste**: if the selection is moved (by the arrow keys or by dragging) beyond the end of the file, or before its start, the file gets longer, in every channel of a set, and the status line says how much; a paste whose points reach past the end (for example with the playhead at the end, the End key) makes the file as long as it needs. One Ctrl+Z takes the move or the paste back together with the extra length. The noise goes with the sines, as it does in a bend. A paste cannot go before the start, because it begins at the playhead. When you drag a selection toward the edge of the picture, the view follows it past the ends of the file, so that a long move can be made in one drag. The room is the same as for a bend.

The noise follows the sines. When the **Noise** layer is on (and the file has noise bands), the noise moves in time together with the sines. This holds for a move (arrow keys or drag), for a bend or stretch with the box, for copy and paste, and for cut (Ctrl+X). Plain Delete removes only the sines and leaves the noise, and transposing never moves noise, because a noise band has a fixed frequency range. A noise band takes part when more than half of its width lies inside the frequency range of the box (or, without a box, of the selection), and the time range is that of the box. Where moved noise lands on other noise the two are **added**, nothing is written over, and the place it came from becomes silent. It is one undo step together with the sines, in every green channel. With the Noise layer off, the noise stays where it is. Dragging a selected point near the edge of the box still moves the selection (see Dragging above); the dots and the squares in the middle of the edges are always the box's own, and the lines between them are only grabbed where there is no selected point under the pointer. Hold Shift to start a new box selection right on a box line. A box that is only a few pixels high or wide, such as one partial or a few frames, is made at least 30 pixels each way when you start to bend, so that the dots do not lie on top of each other.

MAX PARTIALS. A button in the top row of the window, to the right of the two EXPORT buttons. With a selection it thins out the selected partials: you choose how many to keep, and the others are silenced. Without a selection the button is greyed out and does nothing.

Click the button and a panel opens under it. ReaATS has ranked every separate selected partial by its total loudness over its own selected length. A row that holds several unrelated partials, or two separate selected stretches of the same row, counts as separate partials. The slider runs from 1 to the number of partials in the selection, and it starts at the right end: all of them are kept, so nothing is silenced yet. The label shows what you have chosen, for example "keep 40 of 41". The buttons **LOUDEST** and **WEAKEST** choose which partials are kept: the loudest ones, dropping the weakest, or the weakest ones, dropping the loudest. A line under them says what will happen, for example "keeps



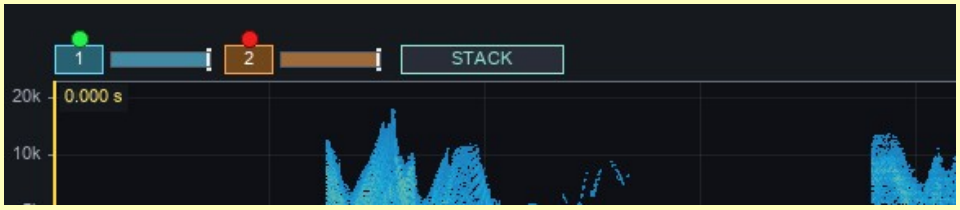
the 40 loudest, silences 1". APPLY, or Enter, silences the others. Esc, CANCEL, or a click outside the panel leaves everything as it was. Up and Down change the number by one, Left and Right by ten, and L and W choose loudest or weakest.

Keeping the loudest cleans the quiet clutter out of a dense analysis. Keeping the weakest does the opposite: it strips away the strongest partials, so that you hear what is left underneath. Only the selected points are ever changed. A partial that continues, unselected, before or after the selection on the same row stays exactly as it was. The change is one undo step, in every channel that has a selection, and Ctrl+Z takes it back.

Stereo and multichannel sounds

A stereo or multichannel analysis gives one .ats file per channel: name_ch1.ats, name_ch2.ats and so on (see the Analysis page). When you open one of them in the editor, with O or with Import .ats File on the Analysis page, ReaATS finds the others and opens the whole set at the channel you picked. The files must have the same frame size, the same number of frames and the same length, as they do when they come from one analysis. If they do not fit together, or one is missing, ReaATS opens the file you chose alone and the status line says why.

Channel buttons. With two or more channels there is an extra row under the header with a button for each channel (1, 2, 3, ...), each in its own colour, and a slider next to it. A button is a switch: click it to turn that channel on or off. A channel that is on is drawn; a channel that is off is not drawn and is not edited. One channel always stays on. A star on a button means that channel has changes that are not saved.



Which channels are edited: the LEDs. Above each channel button is a small LED. Green means the channel is edited together with the other green channels; red means it is left alone. Click the LED to change it. Selecting, delete, transpose and move go only to channels that are green and switched on. A red channel stays visible, it just cannot be changed, and it loses its selection when it turns red. All LEDs start green.

Stack and overlay. The button at the end of the row, or the V key, chooses how the channels that are on are shown. In STACK each channel has its own lane, one under the other, with the same time axis and the same frequency range, so you can compare the channels at a glance; the lanes are named ch 1, ch 2 and so on. In OVERLAY all the channels are drawn in one picture, each in its own colour (the louder the partial, the brighter the colour). Tab shows one channel alone, the next one each time you press it, and after the last channel all channels again.

Visibility sliders. The slider next to a button sets how strongly that channel is drawn, from 20 % to 100 %. Use it in the overlay to push one channel into the background, or in the stack to dim a lane. It only changes the picture, never the edits and never the sound: AUDITION and PRINT always use every channel. A channel that is off counts as 0 %.

Editing channels together. A selection box selects the same time and frequency region in every channel that is on, each channel selecting its own points. Delete, transpose and move (with the arrow keys or by dragging a selected point) then apply to all those channels at once, and the header shows the total. Switch channels off to edit only some of them. An edit in several channels is one step for Ctrl+Z and Ctrl+Y: one press undoes it in every channel it touched.

The stereo image. The ear places a sound from left to right partly by the difference in time between its two ears, which is a difference in phase, most of all at low frequencies. With the stored phase, ReaATS rebuilds every channel with its own waveform, so this difference between the channels of the original comes back. In a test with a stereo tone whose right channel was 0.4 ms late, the rebuilt sound had the same 0.4 ms. Without the phase only the differences in level are left, and the image moves towards the centre. The SPACE modes P-STEREO and VBAP spread the partials over the outputs on purpose and do not keep this relation: use STEREO to keep the image of the original.

Listening, output and printing

AUDITION: playing the sound

The AUDITION button plays the edited sound. ReaATS builds the sound from the current data, the sines from the partials and the noise from the 25 noise bands, writes it to a temporary WAV file and plays that through REAPER's preview. Your project is not touched. The sound is made first, which takes a moment. How long depends on how the sound is made (see below). While it is being made, the AUDITION button blinks, a small bar with the percentage shows under the button, and the status line shows how far it is, so you know the sound will start soon. The sound starts by itself when it is ready. Space stops the sound, and cancels the sound that is still being made. If you change the data while the sound is being made, it is cancelled too, because it would not match any more.



Audition of the selection only. Press A while points are selected and only those points sound: from just before the first selected frame to just after the last, the sines only (no noise), in every channel, through the SPACE setting. Use it to hear what a box or a lasso has caught, one partial or a few, without the rest of the sound. The sound is always made again, by the Python program, and does not replace the sound that AUDITION keeps for replaying. To hear the whole sound with a selection in place, use the AUDITION button.

What you hear follows what you see. With P off you hear only the noise, with N off only the sines, and with both off nothing is played. This lets you listen to the two parts of a sound separately.

The sines come out at the level of the original. The noise is made from the 25 noise bands in the file, as filtered noise.

Listening: AUDITION. AUDITION plays all channels together, from the playhead. REAPER's preview is stereo, so a set with more than two channels is mixed down to stereo for listening: channel 1 goes to the left, the last channel to the right, and the channels in between are spread evenly between them. The level is set so that the mixdown is not louder than one channel would be. A set of exactly two channels is played as it is, channel 1 left and channel 2 right. The status line says "3 channels mixed down to stereo" when this happens. The partials and noise switches count for all channels.

The playhead. AUDITION starts at the playhead. Click anywhere in the picture, on empty space, to put the playhead there. That click also clears the selection. You can also click on the time ruler, the strip with the seconds under the picture, or drag along the ruler to move the playhead. While the sound plays, the playhead moves along the picture. When the sound ends, or when you stop it with Space, the playhead goes back to where you put it, so you can listen to the same place again. The sound starts at the beginning of the analysis frame that holds the playhead, which is at most one frame, 10 ms in the example file, before it. PRINT also starts at the playhead (see below).

Output and level

SPACE: pseudo-stereo and VBAP. Three controls in the top row, to the right of MAX PARTIALS, decide how the sound is spread over the outputs when you AUDITION or PRINT. The same ideas are in SpectralTransfer. The spreading is done per partial (and per noise band), so it needs the Python program that makes the sound; with the built-in synthesis the SPACE button must stay on its first state.

- **SPACE** cycles through three states. The first is named after what comes out: **1 channel** for a single file, **3 channels** for a set of three, and so on. It is the sound as it is made from the channels of the file, and the slider does not apply to it. The other two are **P-STEREO** and **VBAP**.
- **P-STEREO** (pseudo-stereo): partials 1, 3, 5 ... go left and 2, 4, 6 ... go right, and the noise bands do the same by band number. The slider is the **width**: 100 % is the full split, 0 % puts everything in the middle. The sound is made 6 dB louder, because each side only gets half of the partials.
- **VBAP**: every partial gets a position between the outputs and is panned between the two outputs that lie next to it (constant power, so the loudness does not change as a partial sits between two speakers). The width of the position grows with the number of the partial (the first partial is in the middle, the last one is the widest), odd partials go to the left and even ones to the right, and a partial below 100 Hz always stays in the middle so that the bass is steady. The slider is the **spread** (0 %

puts everything in the middle). The **OUT** box is the number of outputs, 2 to 8: click for one more, right-click for one less.

- AUDITION always plays stereo: a VBAP sound is mixed down to two channels for listening. PRINT writes every output as a channel of the WAV and gives the track enough channels.
- With a set of several channels (for example a stereo analysis) the channels are added together and the outputs are made from the partials, so the left and right of the original are not kept in this mode. Put SPACE on its first state to keep them.
- The setting is remembered for the next time.

The level. After copying and pasting, or moving partials onto each other, the sound can become louder than full scale (0 dB). AUDITION and PRINT therefore turn the whole sound down, by one gain for all channels, when its highest peak is above -1 dBFS, so that the peak lands exactly on -1 dBFS. A sound that is quieter is left as it is: it is not turned up. That is on purpose: when you print the sines and the noise separately and mix them in REAPER, each print keeps its own level and the balance between them stays what the analysis made. When PRINT turns a sound down, the status line says by how many dB. AUDITION shows no message of its own (not while it prepares the sound, not while it plays, not when you stop it), only when something goes wrong. The Python program has the options --peak-limit (another level) and --no-limit.

PRINT: putting the result in the project

The PRINT button, or Ctrl+P, turns the current sound into a sample on a track. The sound is made first, as for AUDITION: the PRINT button blinks and a bar with the percentage shows under it. Esc cancels a print that is still being made, and so does a change to the data. ReaATS first saves unsaved changes as name_edit.ats, then renders a permanent WAV in the ReaATS folder of the project (next to the .ats file if the project has not been saved yet), called name_print1.wav, and puts it on the selected track at the edit cursor. If no track is selected, a new track called ReaATS is made. Each press makes a new numbered file, and nothing is overwritten.

Important: PRINT starts at the playhead, like AUDITION. The printed sound runs from the playhead to the end of the sound, so only with the playhead at 0 do you get the whole sound. A click on empty space in the picture, or on the time ruler, moves the playhead, so check it before you print: the small button under PRINT, marked |<, shows where the playhead is ("|< 62.3 s", in amber, when it is not at the start, and "|< start" when it is). Click that button, or press Home, to put the playhead back at 0. End puts it at the end. When you press PRINT with the playhead away from the start, the status line says "preparing the print FROM 62.30 s". The status line says where the print started, for example "printed 1.2 s from 1.00 s". As for AUDITION, the sound starts at the beginning of the analysis frame that holds the playhead. ReaATS remembers the start with the item, and when ReaSnap restores a snapshot the playhead comes back to it.

Printing: PRINT. PRINT keeps all the channels. It makes one multichannel WAV file, one channel of sound for each channel file, from the playhead, and puts it on the selected track. A track needs room for the channels, so ReaATS raises the number of channels of that track to the next even number that is large enough, for example 4 for a sound of 3 channels. REAPER itself must be set up for multichannel use: route the track's channels to where you want them, for example to the outputs of your interface. The item remembers all the files of the set, and ReaSnap brings the whole set back.

The WAV lives in the ReaATS folder inside the project folder, so it moves with the project when you copy the project folder to another computer. If the project had not been saved when you printed, the WAV is next to the .ats file instead and is not part of the project: save the project first, or copy the file by hand.

Saving and exchanging files

Saving

Ctrl+S saves the changes as a new file, name_edit.ats, in the ReaATS folder of the project (next to the original if the project has not been saved yet). If that file exists, the next one is name_edit2.ats, and so on. The original is never overwritten. For a set of channels, every channel is saved, as name_edit_ch1.ats, name_edit_ch2.ats and so on, so that the saved set opens as a set again.

The saved file keeps the phase information (file type 4 when the file has noise bands). Pieces of partials that you did not touch keep the phase of the original. A piece that you changed gets new phases that follow its new frequencies, so the file stays consistent: programs that use the phase, such as pyatsyn, play the edited partials correctly, and Csound reads the file as well. If the opened file had no phase, phases are made for every partial. The largest amplitude and frequency in the file header are recalculated from the edited data.

The EXPORT button, at the top left, or Ctrl+Shift+S, lets you choose the folder and the name of the .ats file. It uses a file dialog (this needs the JS_ReaScriptAPI extension, which most REAPER installations with ReaPack have) or, without it, a box in which you type the full path. The ending .ats is added if you leave it out, and ReaATS asks before it replaces a file that already exists. Use it to export a file for Csound or another program, and to put the file where you want it. Like Ctrl+S, it keeps the phase. For a set of channels you choose one name, and every channel is written from it: choose mysound.ats and you get mysound_ch1.ats, mysound_ch2.ats and so on. ReaATS asks before it replaces any of them.

O opens another file. R loads the file you opened again from disk and drops all changes since.

Where the files go

ReaATS puts everything it makes that belongs to your work in a folder called ReaATS, inside the folder of the REAPER project:

- the analyses, name.ats (name_ch1.ats, name_ch2.ats ... for a sound with several channels);
- the saved copies of your edits, name_edit.ats;
- the prints, name_print1.wav;
- the MIDI files of EXPORT NOTES, name_notes1.mid.

Copy the whole project folder to another computer and these files go with it. The sound that you analysed is not copied: it stays where it was, and the analysis does not need it again. EXPORT .ats FILE is the exception, it puts the file where you choose.

A project that has not been saved yet has no folder. Then ReaATS does what it did before and puts the files next to the sound, and the status line says so. Saving the project later does not move them. Files that exist already are not moved either: the ReaATS folder is used for new analyses, edits, prints and exports.

The settings that ReaSnap keeps with a printed item name the .ats file by its full path. On another computer, open the file from the ReaATS folder by hand (the O key) if the path is not the same.

The temporary files, such as the sounds that AUDITION makes, are never put in the project. They are in the ReaATS_temp folder inside the REAPER resource folder, and ReaATS removes them (see Good to know).

*ReaATS reads and writes **standard ATS files**, the format of Csound's ATS opcodes (ATSread, ATSadd, ATSaddnz, ATSsinnoi and the others). There is no format of its own and no conversion step.*

- ***Into ReaATS:** any .ats file opens, whatever made it: Csound's own analyser (atsa), the ATSA and ATSH programs, pyatsyn, or the analysis in ReaATS. File types 1 to 4 are read (with or without phase, with or without the 25 noise bands), in either byte order. Use the O key, or Import .ats File on the Analysis page.*
- ***Out of ReaATS:** Ctrl+S saves your edits as a new .ats file in the project's ReaATS folder, and EXPORT .ats FILE puts a file where you choose. They are written in the layout that Csound reads (type 4 with phase and noise bands, or type 2 without noise). The analysis in ReaATS makes the same kind of file. A stereo or multichannel sound is one file per channel, name_ch1.ats, name_ch2.ats and so on, because the Csound opcodes read one channel from one file.*
- ***Csound itself is not needed** for any of this. It is only the program that other people's instruments read the files with.*

What has been checked with Csound 7 itself: that Csound reads the files that ReaATS saves (the ATSinfo and ATSadd opcodes), that an analysis from Csound's atsa opens in ReaATS and is saved again without change except the header value for the highest frequency, and that a test sound analysed by ReaATS (three gliding notes) plays back through ATSadd at the same pitches, within 1 Hz, and with the same loudness over time as the original (the loudness curves agreed with a correlation of 0.994). It is one small test sound, not a guarantee for every kind of sound.

Three things to know when you use the files in Csound:

- ***The phase.** ReaATS keeps the phase in the file and uses it when it plays (AUDITION and PRINT), as pyatsyn does. Csound's ATSadd does not use the stored phase: its oscillators start with a phase of their own. The sound has the same partials and the same loudness, but a waveform printed from Csound will not*

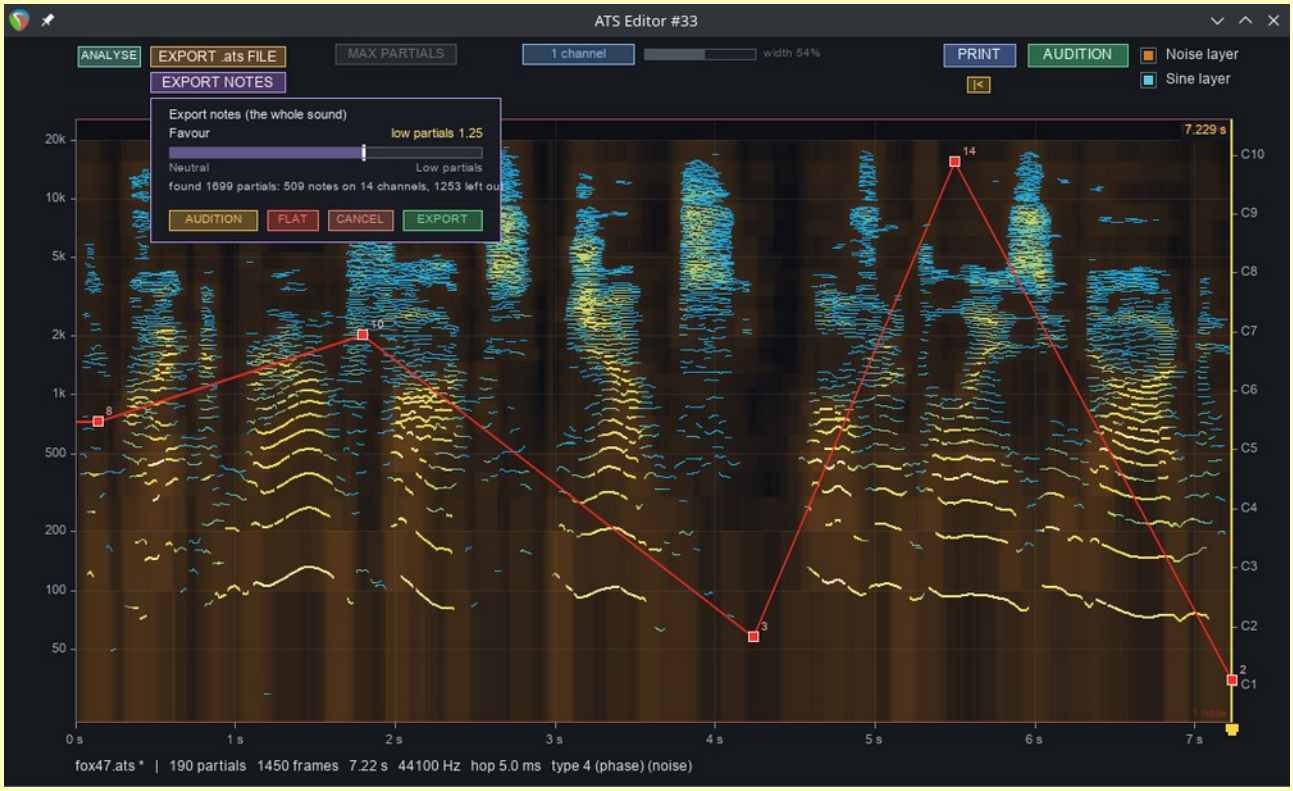
be sample for sample the same as one printed from ReaATS.

- ***Rows are not always single partials.*** *With Share rows between tracks on (the default) one row of the file can hold several partials one after the other, with a silent frame between them, which keeps the file many times smaller. Played in ATSadd the result is the same as without sharing (the loudness curves of a shared and an unshared analysis of the same sound agreed to 0.9998). But an instrument that reads one row with ATSread, or that uses the number of the row as the number of a partial, will see the row jump from one partial to the next. If your Csound instrument depends on partial numbers, switch Share rows between tracks off before analysing.*
- ***Edits can add rows.*** *A move or a paste can need a new row, so a file that you edited may have more partials than the original. Tell ATSadd the number of partials of the edited file, which the line at the bottom of the window shows.*

Export notes: from sound to score

EXPORT NOTES turns the partials of a sound into a score. Each partial that sounds for long enough becomes one MIDI note on a channel of its own, and the way the partial moves, its pitch drifting and its level rising and falling, becomes pitch bend and pressure. The result is a MIDI item in REAPER, and a MIDI file next to it, that any MPE instrument can play. With this button ReaATS is not only an editor for partials but also a generator of scores: the rhythm, the chords and the melodic gestures of a sound, played by an instrument that you choose.

You decide how many notes may sound at each moment by drawing a red line over the picture. With few notes the score has the shape of the sound: its timing, its phrases and its melodic movement. With more notes it gets closer to the sound itself.



How to use it

1. Prepare. Open or analyse a sound. To use only a part of it, a word or a phrase, select that part first (a box, a lasso, or Ctrl+A for everything). Without a selection the whole sound of the channel that is shown is used. Select the track that has your MPE instrument on it. The notes are put on the selected track.

2. Click EXPORT NOTES, under the EXPORT .ats FILE button. A small panel opens, and a red line appears over the picture.

3. Shape the score with the red line and the Favour slider (see below), and press AUDITION to hear it while you work.

4. Click EXPORT, or press Enter. Esc or CANCEL closes the panel and changes nothing.

Two things are made. A MIDI item with the notes is put on the selected track, at the edit cursor. If no track is selected, a new track called "ReaATS notes" is made for it. And a MIDI file called name_notes1.mid is written in the ReaATS folder of the project (next to the sound if the project has not been saved yet). The number counts up, so no earlier export is overwritten. The status line says how many notes were made, on how many channels, and where they went.

The red line: how many notes at each moment

The top of the picture is 15 notes and the bottom is 1 note. The red line says how many notes may sound at the same moment. At the start it is a flat line at 2 notes, so that a plain export already gives you the melody of the sound.

- **Click in the picture** to add a point to the line. **Drag** a point to move it up and down (the number of notes) and sideways (the time). **Right-click** on a point to remove it.
- The two ends of the line are points too. They sit at the start and the end of the selection, or of the whole sound.
- **FLAT** removes your points and gives you the flat line at 2 notes again.
- A number next to each point shows how many notes it allows.

Where the line is high, many partials sound together; where it is low, only the strongest. When the number falls, the notes that no longer have room are cut short, and the strongest partials keep their places. A note is never taken from another one, and a stretch that would be shorter than three frames is left out. The line in the panel ("found 212 partials: 11 notes on 11 channels, 201 left out") counts what your drawing gives.

You can use the line as a dramatic device: start low, at 2 notes, so that the listener hears only intention, and let it rise to 8 or more, so that the words appear out of the music. Or the other way round: let a clear text dissolve into a melody.

Favour: which partials are chosen

Partials are chosen by their energy: the strongest first. In speech, and in many instruments, the loudest partials are the high ones around the formants, so the fundamental and the low harmonics may be left out. The Favour slider goes from Neutral (strongest first) to Low partials. The further to the right, the more the choice is pulled to the low partials: at 1 a partial an octave lower counts twice as much, at 2 four times as much. It starts at 0.5 and is remembered. The low partials carry the pitch and the melody; the high ones carry the vowels and consonants. Try the same sound with different settings, and listen.

Left and Right on the keyboard change Favour in small steps.

AUDITION

The AUDITION button in the panel (or the A key) lets you hear the notes before you export them. ReaATS puts a temporary MIDI item on the selected track, at the edit cursor, and plays it in a loop (REAPER's Repeat is switched on, and the time selection is set to the item). While you draw on the red line or move the Favour slider, the item is rewritten, so you hear the changes at once. The button turns into STOP. STOP, Esc, CANCEL, EXPORT, or stopping REAPER's transport yourself ends the audition: the temporary item is deleted, and Repeat, the time selection and the edit cursor are put back as they were.

AUDITION needs a selected track with an instrument on it. Without a selected track the status line says so.

What is in the notes

What you hear	What it comes from
A note	One partial, from the moment it starts to sound until the moment it stops, or the part of it for which the red line allows a place. A partial that goes silent and comes back is two notes.
The pitch of the note	The semitone nearest to the average pitch of the partial.
Pitch bend	How far the partial is from that semitone, at every moment. This keeps what makes an ATS analysis special: inharmonic partials, vibrato, glides and the intonation of a voice.
Pressure	The level of the partial, at every moment. It is sent in dB, so a fade sounds even. 127 is the loudest partial of the file, 0 is 60 dB below that.
Velocity	The level of the partial at the moment the note starts.
Channel	Each note sounds on a channel of its own, 2 to 16. Channel 1 is the MPE master channel.

At the start of the file the MPE set-up is sent: the MPE zone on the master channel, and a pitch bend range of 48 semitones on every member channel. This is the MPE standard.

What the instrument must do

An MPE instrument is needed, and it must do two things with what it receives. It must follow pitch bend per note, with a range of 48 semitones. If your instrument has its own bend range setting, it must be 48. And it must connect pressure to level: in many instruments this has to be assigned, because pressure may be set to control something else, such as brightness. If the notes play but do not follow the pitch gestures, check the bend range. If they follow the pitch but all sound equally loud, check what pressure controls.

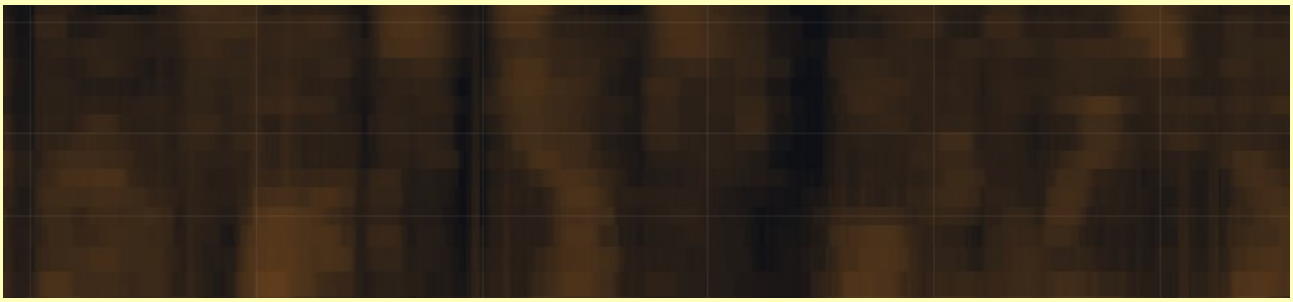
A sine-based instrument is the closest match to the sound that ReaATS plays itself. Other instruments give you something different: the partials of a voice played by strings, a pad or a plucked sound are a way to make a score from the voice.

What is not in the notes

- The noise bands are not exported. Breath, friction and attack noise are missing.
- The phase is not exported. Notes begin on their own.
- Only partials that sound for at least three frames are used, and only as many as the red line allows. The result is a sketch of the sound, not a copy of it.
- A set with several channels (see the chapter on stereo and multichannel sounds) exports the channel that is shown. To export another channel, switch to it and export again.

Removing noise

ReaATS is made for creative work: to take a sound apart, change it and make something new from it. Cleaning a recording is not what it was built for, but it comes with the way it works, and it is useful to know. An analysis splits a sound in two: the partials, the sine lines you see in the picture, and the noise, the orange layer behind them. A sound that is made again from the partials alone is a sound without the noise. That also makes it possible to remove noise from sounds that have clear pitched parts, such as a voice, a wind instrument, a string, a bowed or plucked note, or a recording with a steady hum. It is not a filter: the clean sound is built again from the analysis, so you are listening to a resynthesis. This chapter shows how to do it with what ReaATS has now.



Step 1: analyse so that the noise is left out of the partials. On the Analysis page three options decide how much of the noise the analysis follows as if it were music:

- Raise the Amplitude threshold, for example from -60 dB to -50 or -40 dB. Hiss is made of many quiet peaks, and a higher threshold does not follow them. The partials that remain are the louder ones. A higher threshold also makes the analysis much faster.
- Raise the Minimum segment length, for example from 3 to 6 or 10 frames. Noise produces short, flickering tracks; this drops every track shorter than that.
- Keep Optimise on, so that the quiet and short tracks are pruned afterwards.

Set the Lowest frequency a little below the lowest note you want to keep (for a voice 70 Hz instead of 20 Hz), so that the window is not longer than it needs to be. Leave Store the noise bands on: you need the noise bands to hear the difference and to decide how much to keep.

Step 2: listen with and without the noise. Open the result and press A for AUDITION. Then switch the Noise layer off (the checkbox at the top right, or the N key) and press A again. This is the denoised sound. Because the noise layer is shown in orange you can also see what you are taking away. Switching the layer on and off while the same sound plays again is the quickest way to judge it: if the sound is still alive without the noise, you are almost done.

Step 3: remove what is left in the partials. Even with a high threshold some noise stays in the sine layer. You see it as faint, short, scattered lines, often in the high frequencies, and as long straight lines that never change: a hum at 50 or 60 Hz and its multiples, or the whine of a fan. Drag a box around them and press Delete. Zoom in with the mouse wheel to separate them from the notes you want. The box selects only the points inside it, so a note that crosses the box is not deleted as a whole. With a stereo or multichannel sound the box and Delete apply to every channel whose LED is green, and Ctrl+Z undoes it in all of them at once. AUDITION again after each step: the data is the only thing you change, the recording is never touched.

Step 4: print it. With the Noise layer switched off, press Ctrl+P or the PRINT button. The clean sound is made as a WAV file in the ReaATS folder of the project and put on the selected track (or a new track) at the edit cursor as a new item. The original item stays where it is: put the new one on another track, or mute the original, to compare. Put the ReaATS playhead at 0 if you want the whole sound. ReaATS saves your edits as name_edit.ats first, so you can come back to them, and ReaSnap brings the settings back (see PRINT: putting the result in the project).

What to expect.

- The result is a resynthesis. What is not a partial is gone: breath, the consonants of speech, the pluck of a string, the click of a key and other noisy parts of the sound that belong to it. If the sound feels thin, dry or "watery", the noise layer is carrying too much of the character to be left out. The noise

layer is all or nothing at present: there is no control for a part of its level, so you cannot yet keep it 12 dB down.

- Noise that sits under a partial is not removed from that partial. It is only removed when the noise is a separate track, or in the noise bands.
- A threshold that is too high removes the quiet end of a note, for example the decay of a long tone. Lower it a little, and make a new analysis, when notes end too early.
- The quality depends on the analysis. Pitched sounds with steady noise work best. Dense music with many instruments, or a sound that is mostly noise, gives a messy picture that is hard to clean by hand.
- A higher threshold removes more of the noise and makes the analysis faster.

All controls

Control	What it does	
Left drag	Select with a box. With Shift: add to the selection.	
Ctrl + Shift + left drag	Select with a lasso: draw a loop round the points. It adds to the selection (Esc first to start over).	
Ctrl+A (Cmd+A on a Mac)	Select the whole sound, with the box all round it. Esc clears it.	
Drag a dot of the box, or an edge between the dots	Bend the selected points in frequency (up and down) and in time (sideways: stretch or squeeze). Ctrl+click on an edge, away from its square handle: a new dot. Right-click on a dot: remove it. Ctrl+Shift+drag on an edge: arch it. Ctrl while dragging a dot or an edge (also when held from the press): one direction only. Shift: fine.	
Enter, Esc or Ctrl+R	Apply the bend of the box, cancel it, put the box flat.	
MAX PARTIALS button (top row; greyed out with no selection)	Asks how many of the SELECTED partials to keep: positive keeps that many loudest, negative that many weakest. Applies right away, only to what was selected.	
Left drag on a selected point	Move the selection. With Ctrl: one direction only (the time or the frequency stays). With Shift: fine movement. A crosshair shows where it goes.	
Click on empty space	Clear the selection.	
Right or middle drag	Pan.	
Mouse wheel	Zoom in time. With Ctrl: zoom in frequency. With Shift: scroll in time.	
Delete	Silence the selected points.	
Up and Down	Transpose by a semitone. Shift: 10 cents. Ctrl: an octave.	
Left and Right	Move one frame in time. Ctrl: ten frames.	
Ctrl+C, Ctrl+X, Ctrl+V	Copy, cut, paste the selected points. The paste starts at the playhead.	
Ctrl+Z, Ctrl+Y	Undo, redo.	
Ctrl+S	Save as name_edit.ats.	
EXPORT .ats, Ctrl+Shift+S	Export an .ats file: choose the folder and the name.	
ANALYSE	Open the Analysis page.	
A	With a selection: play the **selection only** (see AUDITION: playing the sound). Without one: play the edited sound from the playhead, like the AUDITION button.	
AUDITION button	Play the edited sound, from the playhead.	

Control	What it does	
Click on the picture, or click or drag on the time ruler	Set the playhead.	
Space	Stop the sound.	
Ctrl+P, PRINT	Put the edited sound on a track, from the playhead.	
P, N, the checkboxes Sine layer and Noise layer	Show or hide the partials, the noise. This also decides what is heard.	
F, right-click	Fit the view (the whole file).	
Home, the	< button under PRINT	Put the playhead at the start of the sound. The button shows where the playhead is. End: at the end.
Tab	Show one channel alone, the next each time; after the last, all channels again.	
V, the STACK / OVERLAY button	Channels under each other, or in one picture.	
The channel buttons and sliders	Switch a channel on or off; how visible it is (20-100 %).	
O, R	Open a file, reload the file.	
Esc	Clear the selection.	

Good to know

- Long, dense analyses used to be enormous, because every track got a row of its own in the file. With Share rows between tracks (on by default) the file only has as many rows as there are tracks at the same moment, so a long sound is no longer a problem in the file. The editor holds all the numbers of the analysis in memory, about twice the size of the .ats files. Opening a new analysis first lets go of the one that is open. On Linux, ReaATS looks at how much memory the computer has free before it loads, and if the analysis does not fit it does not load it and tells you how much it needs. In that case analyse again with a higher Amplitude threshold (-50 or -40 dB) or a higher Lowest frequency, or analyse one channel at a time. The same measures make the analysis faster.
- The file stores one value per frame, and the frames are evenly spaced. In the example file they are 10 ms apart. Moves in time snap to whole frames.
- ReaATS never overwrites the original .ats file. Saved changes always go to a new file.
- The temporary WAV files of AUDITION are kept in the temporary folder, numbered. ReaATS keeps only the last two. All of them are deleted when the editor closes, and files that an earlier session left behind (after a crash, for example) are deleted at the next start once they are more than six hours old.
- Keep the original .ats file. A saved file is a new file, but the phases of changed partials are calculated and not measured.